

Unit 06: User Interface Design

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Objective

After studying this unit, you will be able to:

- user interface design
- golden rules
- interface design process
- interface design activities

Introduction

The User Interface (UI) design principles are the methods/processes for designing the front end view of a software application so that the client or user may easily interact/use it without risk. If the user interface (UI) of a software application is appealing, clear, intelligible, and responsive, the software application will be very useful. Simply said, UI design principles give a user interface/view for a software product with a focus on look and feel and style, i.e. it is an evolutionary approach that provides maximum usability and the best user experience.

The process of developing or constructing the interfaces through which a user can communicate with a computer is known as user interface design. It is the software engineer who is in charge of creating the user interface. The software engineer ensures that the user interface is simple to comprehend, achieves goals, is enjoyable to use, and is encouraging and forgiving.

The user interface is a device or a programme (application programme) that allows a user to communicate with a computer and tell it what they want to accomplish or get from it. When consumers use a computer, the user interface is what they see. Whatever you see on the computer screen, as seen in the image below, is a user interface.

6.1 Different User Interface Design Principles

Because the popularity of a generated software application is based on the software product's user interface design, user interface design principles can help to improve the quality of the software application's user interface. The Graphical User Interface (GUI) or command line interface of the

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software application is the focus of these principles. The user interface should be simple, meaningful, and easy to grasp. There are several UI design principles to consider.

Given below are user interface design principles:

The Concept of Structural

The Concept of Simplicity

The Concept of Visibility

The Concept of Feedback

The Concept of Tolerance

The Concept of Reusability

1. The Concept of Structural

This is the first stage in designing the user interface for a software application, and it deals with the overall UI architecture. It offers the software application's structural data, as well as combining application-related data and separating undesirable items. This notion ensures that the software application is well-organized (meaningful), innovative (technology advancement), intelligible (software application purpose), and beneficial (satisfy certain criteria).

2. The Concept of Simplicity

This technique implies that the user interface of a software application should be simple, long-lasting, and user-friendly, with user control over the interface, sustainability, complexity clarification, accurate throughput response, maximum usability, and user language. We anticipate our users to make UI mistakes, justify our data, avoid developing a busy interface, and explain the rules for designing a user interface for a software programme. It will always assist in reducing the amount of activities required to perform all of the actions on the UI, as well as safeguarding the users' data in the software application.

3. The Concept of Visibility

This concept of UI design technique involves the visibility of user interface for a software application. It mainly deals with graphical user interface i.e. the alignment is correct or not, spelling checking, position of logo or banners, consistency, easy to navigate with status, colour & brightness of UI and without hesitation to the user. So that the look & feel of the interface should be perfect, clarity, progressive disclose, transparency and error preventive. We should emphasis on the performance issue of the UI.

4. The Concept of Feedback

This principle indicates the enhancement of user interface view depends of feedback. It always welcomes to change or modify the UI depends upon the user or client reviews. So the designer always analyse the process, view, any ambiguity issues, bugs, technological advancement, change of conditions and to make more user friendly of user interface of the software product. For users every action on the UI returns a meaningful and clear reaction or feedback.

5. The Concept of Tolerance

This is one of the concepts of user interface design for the software application which deals or affects the budgeting of the application. The UI designer will design the interface that will be always flexible and tolerant, so it reducing the cost of rework of user interfaces change. For any stage of the application the UI should be tolerable in sense.

6. The Concept of Reusability

This is the last user interface design technique. The user interface view should be reusable within the software application, according to the designer. Some internal or external components, component behaviour, and application consistency should be reused within it. So that the user does not have to remember the application's interconnected process flow on the user interface. It gives the user several shortcut keys to make working on that easier.

The user interface design mainly deals with graphical user interface (GUI) design, which involves multiple characteristics like graphics of the UI, Icons, Windows, Menus, font size, color, alignment, space between lines, etc. here we choose some specific techniques for the design of UI for the Software application.

The main idea behind user interface design to make it so comfortable for a user then, the user easily interacts with the software product. The tag line for the design of user interface process will be “The best User Interface design can attracts more customers”.

6.2 Categories of User Interface

- Command-line Interface
- Menu-Based Interface
- Graphical User Interface

1. Command Line Interface

The command-line interface was the only mode of interaction with the computer in the starting days of computing. With the command-line interface, the user has to communicate with the computer using the textual instructions or we say commands. For example, if one needs to delete a file from the system. The command would be : `del c:\file.txt`

But, this kind of user interface is not user-friendly as it is hard to learn such commands, even they can be error-prone and unforgiving if an error occurs. Such an interface may be irritating. So, the command line interface was not meant for the casual user but the experienced users generally prefer command-line interface. Example: Unix operating system

Advantages

Many and easier to customizations options

Typically capable of more important tasks

Disadvantages

Relies heavily on recall rather than recognition

Navigation is often more difficult.

A text-based command line interface can have the following elements:

Command Prompt - It is text-based notifier that is mostly shows the context in which the user is working. It is generated by the software system.

Cursor - It is a small horizontal line or a vertical bar of the height of line, to represent position of character while typing. Cursor is mostly found in blinking state. It moves as the user writes or deletes something.

Command - A command is an executable instruction. It may have one or more parameters. Output on command execution is shown inline on the screen. When output is produced, command prompt is displayed on the next line.

2. Menu Based Interface

It is somewhat easy than the command line interface as users interacting with menu-based interface ‘do not have to learn the command name’, nor it had to put efforts in writing the commands. As here, the user does not have to type the commands the syntax error is automatically avoided.

A very popular example of a menu-based interface is ATM. A user uses menu buttons to select the options from the menu.

3. Graphical User Interface (GUI)

Over a period of time, there has been a tremendous evolution in the computer’s user interface. As today’s computer have a high-resolution screen with pointing devices (mouse). The modern-day computer’s interface has windows, scroll bars, text boxes, buttons, icons, menus and to pick and point them we have a mouse.

Graphical user interfaces are easy to learn as compared to the command-line interface. GUI provides multiple windows to the user at the same time, to interact with the system and even the user is allowed to switch between the windows.

GUI Characteristics

Characteristics

Descriptions

GUI Elements

GUI provides a set of components to interact with software or hardware.

Every graphical component provides a way to work with the system. A GUI system has following elements such as:

Window - An area where contents of application are displayed. Contents in a window can be displayed in the form of icons or lists, if the window represents file structure. It is easier for a user to navigate in the file system in an exploring window. Windows can be minimized, resized or maximized to the size of screen. They can be moved anywhere on the screen. A window may contain another window of the same application, called child window.

Tabs - If an application allows executing multiple instances of itself they appear on the screen as separate windows. Tabbed Document Interface has come up to open multiple documents in the same window. This interface also helps in viewing preference panel in application. All modern web-browsers use this feature.

Menu - Menu is an array of standard commands, grouped together and placed at a visible place (usually top) inside the application window. The menu can be programmed to appear or hide on mouse clicks.

Icon - An icon is small picture representing an associated application. When these icons are clicked or double clicked, the application window is opened. Icon displays application and programs installed on a system in the form of small pictures.

Cursor - Interacting devices such as mouse, touch pad, digital pen are represented in GUI as cursors. On screen cursor follows the instructions from hardware in almost real-time. Cursors are also named pointers in GUI systems. They are used to select menus, windows and other application features.

Application specific GUI components

Application Window - Most application windows uses the constructs supplied by operating systems but many use their own customer created windows to contain the contents of application.

Dialogue Box - It is a child window that contains message for the user and request for some action to be taken. For Example: Application generates a dialogue to get confirmation from user to delete a file.

Text-Box - Provides an area for user to type and enter text-based data.

Buttons - They imitate real life buttons and are used to submit inputs to the software.

Radio-button - Displays available options for selection. Only one can be selected among all offered

Check-box - Functions similar to list-box. When an option is selected, the box is marked as checked. Multiple options represented by check boxes can be selected.

List-box - Provides list of available items for selection. More than one item can be selected.

Advantages

Less expert knowledge is required to use it.

Easier to Navigate and can look through folders quickly in a guess and check manner.

The user may switch quickly from one task to another and can interact with several different applications.

Disadvantages

Typically decreased options

Usually less customizable Not easy to use one button for tons of different variations.

6.3 Golden rule

The user interface (UI) is a critical part of any software product. When it's done well, users don't even notice it. When it's done poorly, users can't get past it to efficiently use a product. To increase the chances of success when creating user interfaces, most designers follow interface design principles. Interface design principles represent high-level concepts that are used to guide software design. In this article, I'll share a few fundamental principles. These are based on Jakob Nielsen's 10 Usability Heuristics for UI Design, Ben Shneiderman's The Eight Golden Rules of Interface Design, and Bruce Tognazzini's Principles of Interaction Design. Most of the principles are applicable to any interactive systems — traditional GUI environments (such as desktop and mobile apps, websites) and non-GUI interfaces (such as voice-based interaction systems).

- Place users in control of the interface
- Make it comfortable to interact with a product
- Reduce cognitive load
- Make user interfaces consistent

1. Place users in control of the interface

Good UIs instill a sense of control in their users. Keeping users in control makes them comfortable; they will learn quickly and gain a fast sense of mastery.

Make actions reversible – be forgiving

This rule states that the user should always be able to rapidly return to their previous state. This lets customers to experiment with the product without fear of failure – knowing that mistakes can be readily undone encourages users to try out new things. On the other hand, requiring a user to be extremely cautious with every step they take results in a slower exploration and a nerve-wracking experience that no one wants.

Perhaps the most common GUIs where users have the 'Undo/Redo' option are text and graphics editors. While writing text or creating graphics, 'Undo' lets users make changes and go back step-by-step through changes that were made. 'Redo' lets users undo the undo, which means that once they go back a few steps, they are able to move forward through their changes again.

'Undo' can be extremely helpful when users choose system function by mistake. In this case, the undo function serves as an 'emergency exit,' allowing users to leave the unwanted state. One good example of such emergency exits is Gmail's notification message with an undo option when users accidentally delete an email.

Create an easy-to-navigate interface

Navigation should always be clear and self-evident. Users should be able to enjoy exploring the interface of any software product. Even complex B2B products full of features shouldn't intimidate users so that they are afraid to press a button. Good UI puts users in their comfort zone by providing some context of where they are, where they've been, and where they can go next:

Provide visual cues. Visual cues serve as reminders for users. Allow users to navigate easily through the interface by providing points of reference as they move through a product interface. Page titles, highlights for currently selected navigation options, and other visual aids give users an immediate view of where they are in the interface. A user should never be wondering, "Where am I?" or "How did I get to this screen?"

Predictability: Users should be provided with cues that help them predict the result of an action. A user should never be wondering, "What do I need to press in order to do my task?" or "What is this button for?"

Provide informative feedback – be acknowledging

Feedback is typically associated with points of action — for every user action, the system should show a meaningful, clear reaction. A system with feedback for every action helps users achieve their goals without friction.

UI design should consider the nature of interaction. For frequent actions, the response can be modest. For example, when users interact with an interactive object (such as a button), it's essential to provide some indication that an action has been acknowledged. This might be something as

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simple as a button changing color when pressed (the change notifies the user of the interaction). The lack of such feedback forces users to double-check to see if their intended actions have been performed.

For infrequent and significant actions, the response should be more substantial. For example, when filling out a password field in the signup form, good UI might inform users of the requirements for their password.

Show the visibility of system status

Users are much more forgiving when they have information about what is going on and are given periodic feedback about the status of the process. Visibility of system status is essential when users initiate an action that takes some time for a computer to complete. Users don't like to be left seeing nothing on the device screen while the app is supposed to be doing something. The use of progress indicators is one of the subtle aspects of UI design that has a tremendous impact on the comfort and enjoyment of users.

Good UI can comfort users by showing progress while the system is completing a task. Drop box is indicating the status of a document upload: the current progress and the amount of time left.

Accommodate users with different skill levels

Users of different skill levels should be able to interact with a product at different levels. Don't sacrifice expert users for an easy-to-use interface for novice or casual users. Instead, try to design for the needs of a diverse set of users, so it doesn't matter if your user is an expert or a newbie.

Adding features like tutorials and explanations is extremely helpful for novice users (just make sure that experienced users are able to skip this part).

Once users are familiar with a product, they will look for shortcuts to speed up commonly-used actions. You should provide fast paths for experienced users by enabling them to use shortcuts.

2. Make it comfortable for a user to interact with a product

Eliminate all elements that are not helping your users

Interfaces shouldn't contain information that is irrelevant or rarely needed. Irrelevant information introduces noise in UI –it competes with the relevant information and diminishes its relative visibility. Simplify interfaces by removing unnecessary elements or content that does not directly support user tasks. Strive to design UI in a way that all information presented on the screen will be valuable and relevant. Examine every element and evaluate it based on the value it delivers to users.

A good example of an app that follows the 'less is more' approach by avoiding overloading the interface with content or features is iA Writer.

The interface of iA Writer app is a clean typing sheet with no distractions. It allows users to focus on what they're writing and hides everything else.

Don't ask users for data they've already entered

Don't force users to have to repeat data they've previously entered. Users are easily annoyed by tedious data-entry sequences, especially when they have provided all the required information before. Good UI performs a maximum of work while requiring a minimum amount of information from users.

Avoid jargon and system-oriented terms

When designing a product, it's important to use language that is easy to read and understand. The system should speak the user's language, with words, phrases, and concepts familiar to the user, rather than jargon or system-oriented terms.

Apply Fitts's Law to interactive elements

Fitts Law states that the time to acquire a target is a function of the distance to and size of the target. This means that it's better to design large targets for important functions (big buttons are easier to interact with).

It's also important to remember that the time required to acquire multiple targets is the sum of the time to acquire each. Thus, when working on UI design, to increase the efficiency of an interaction, try to not only reduce distances and increase target sizes, but also reduce the total number of targets that users must interact with to complete a given task.

Design accessible interfaces

When we design products it's important to remember that a well-designed product is accessible to users of all abilities, including those with low vision, blindness, hearing impairments, cognitive impairments, or motor impairments. Good UI is accessible UI because improving your product's accessibility enhances the usability for all groups of users.

Color is one of the elements of an interface that has a strong impact on accessibility. People perceive color differently — some users can see a full range of colors, but many people can only make out a limited range of colors. Approximately 10 percent of men and one percent of women have some form of color blindness. When designing interfaces, it's better to avoid using color as the only way to convey information. Anytime you want color to convey information in the interface, you should use other cues to convey the information to those who cannot see the colors.

Use real-world metaphors

Using metaphors in UI design allows users to create a connection between the real world and digital experiences. Real-world metaphors empower users by allowing them to transfer existing knowledge about how things should look and work. Metaphors are often used to make the unfamiliar familiar. Take the recycle bin on your desktop, which holds deleted files, as an example – it's not a real trash bin, but it's visually represented in a way that helps you understand the concept more easily.

Good metaphors generate a strong connection to past experiences from the real world in users' minds. The recycle bin icon on Macs is similar to an actual bin, and it shows whether it has files in it.

When choosing a metaphor for UI, select the one that will enable users to grasp the finest details of the conceptual model. For example, when asking for credit card details for payment processing, you can reference a real-world physical card as an example.

Engineer for errors

Errors are inadvertent in the user journey. Bad error handling paired with useless error messages can fill users with frustration and lead them to abandon your app. A well-crafted error message, on the other hand, can turn a moment of frustration into a moment of conversion. An effective error message is a combination of explicit error notification together with hints for solving the problem.

Even better than writing good error messages is having UI design that prevents a problem from occurring in the first place. Try to either eliminate error-prone conditions or check for them and present users with a confirmation dialog before they commit to the action. For example, Gmail prompts you when you forget to insert an attachment.

The best designs have excellent error recovery while trying to prevent users from making those errors in the first place. Error prevention in Gmail shows a pop-up if users forget to insert an attachment after referencing one.

Protect a user's work

Ensure that users never lose their work. Users should not lose their work as a result of an error on their side (i.e. accidentally refresh a web page with a form that has user input), a system error, problems with an internet connection, or any other reason other than those that are completely unavoidable, like an unexpected power loss.

3. Reduce cognitive load

Cognitive load is the amount of mental processing power required to use a product. It's better to avoid making users think/work too hard to use your product.

Chunking for sequences of information or actions

In 1956, psychologist George Miller introduced the world to the theory of chunking. In his works, Miller says the human working memory can handle seven-plus-or-minus two “chunks” of information while we're processing information.

This rule can be used when organizing and grouping items together. For example, if you're UI forces users to enter telephone numbers without normal spacing it can result in a lot of incorrectly-captured phone numbers. People cannot typically scan clusters of ten or more digits to discover errors. That's exa Reduce the number of actions required to complete a task

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When designing a user interface, strive to reduce the total number of actions required from a user to achieve the goal. It's worth remembering the three-click rule, which suggests the user of a product should be able to find any information with no more than three mouse clicks.

Recognition over recall

One of the Jakob Nielsen's 10 usability heuristics advises promoting recognition over recall in UI design. Recognizing something is much easier than recalling it because recognition involves more cues in our brain (cues spread activation to related information in memory, and those cues help us remember information).

Designers can promote recognition in user interfaces by making information and functionality visible and easily accessible. Visual aids, such as tooltips and context-sensitive details, also help support users in recognizing information. Ctlly why phone numbers are broken up into smaller pieces

Promote visual clarity

Good visual organization improves usability and legibility, allowing users to quickly find the information they are looking for and use the interface more efficiently.

When designing layouts:

- Avoid presenting too much information at one time on the screen. Construct a grid system design to avoid visual clutter.
- Remember the principle 'form follows function.' Make things look like they work.
- Apply the general principles of content organization such as grouping similar items together, numbering items, and using headings and prompt text.

4. Make user interfaces consistent

Consistency is an essential property of good UI—consistent design is intuitive design. Consistency is one of the strongest contributors to usability and Learnability. The main idea of consistency is the idea of transferable knowledge — let users transfer their knowledge and skills from one part of an app's UI to another and from one app to another app.

Visual consistency (style)

Users should never question the integrity of a product. The same colors, fonts, and icons should be present throughout the product. Be sure to always reference your design system manager to ensure you don't change visual styles within your product for no apparent reason. For example, a Submit button but Avoid using different styles for elements on different pages of the site. Users should not have to wonder whether a transformed button like this example means the same thing.

Functional consistency (behavior)

Consistency of behavior means the object should work in the same way throughout the interface. The behavior of interface controls, such as buttons and menu items, should not change within a product. Users don't want surprises or changes in familiar behavior — they become easily frustrated when things don't work. This can inhibit learning and stop users from feeling confident about consistency in the interface. Do not confuse your user — keep actions consistent by following "The principle of least surprise," to have the interface behave the way users expect it to.

Consistent with user expectations

People have certain expectations about the apps/websites they use. Designing your product in a way that contradicts a user's expectations is one of the worst things you can do to a user. It doesn't matter what logical argument you provide for how something should work or look. If users expect it to work/look a different way, you will face a hard time changing those expectations. If your approach offers no clear advantage, go with what your users expect.

Follow platform conventions. Your product should be consistent with standards dictated by platform guidelines. Guidelines ensure that your users can understand individual interface elements in your design.

Don't reinvent patterns. For most design problems, proper solutions already exist. These solutions are called patterns. Popular patterns become conventions and the majority of users are familiar with them. Not taking this solution into account and continuing to design your own solution can lead to challenges for users. In most cases, breaking design conventions results in a frustrating user experience — you'll face usability problems not necessarily because your solution will be wrong, but because users won't be familiar with it.

Don't try to reinvent terminology. Avoid using new terms when there are words available that users already know. Users spend most of their time in other apps and on other sites, so they have certain expectations about naming. Using different words might confuse them. Tton on one page of your site should look the same on any other page.

6.4 Interface Design Models

Four different models come into play when a user interface is to be analyzed and designed

User model: a profile of all end users of the system Users can be categorized as:

Novices: No syntactic and little semantic knowledge of the system.

Knowledgeable, intermittent users: reasonable knowledge of the system.

Knowledgeable, frequent users: good syntactic and semantic knowledge of the system

Design model: a design realization of the user model that incorporates data, architectural, interface, and procedural representations of the software.

Mental model (system perception): the user's mental image of what the interface is. The user's mental model shapes how the user perceives the interface and whether the UI meets the user's needs.

Implementation model: the interface "look and feel of the interface" coupled with all supporting information (documentation) that describes interface syntax and semantics.

6.5 Interface Design Process

The analysis and design process for UIs is iterative and can be represented using a spiral model.

The user interface analysis and design process encompasses four framework activities:

1. User, task and environment analysis and modeling.
2. Interface design
3. Interface construction (implementation)
4. Interface validation

User, task and environment analysis and modeling

In this phase focus is based on the profile of users who will interact with the system, i.e. understanding, skill and knowledge, type of user, etc, based on the user's profile users are made into categories. From each category requirements are gathered.

The analysis of the user environment focuses on the physical work environment. Among the questions to be asked

- Where will the interface be located physically?
- Will the user be sitting, standing, or performing other tasks unrelated to the interface?
- Does the interface hardware accommodate space, light, or noise constraints?
- Are there special human factors considerations driven by environmental factors?

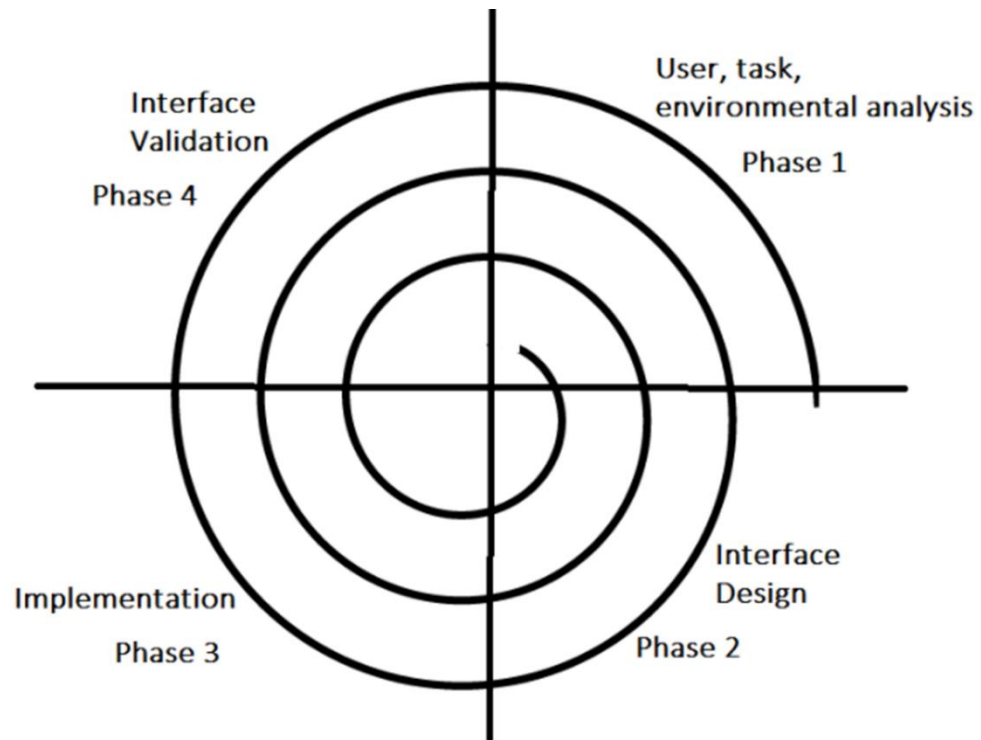
Interface design analysis: The people (end-users) who will interact with the system through the interface;

The tasks that end-users must perform to do their work,

The content that is presented as part of the interface,

The environment in which these tasks will be conducted

Interface design process



User Analysis

- User Interviews
- Sales input
- Marketing input
- Support input

Task Analysis and Modeling: The goal of task analysis is to answer the following questions:

- What work will the user perform in specific circumstances?
- What tasks and subtasks will be performed as the user does the work?
- What specific problem domain objects will the user manipulate as work is performed?
- What is the sequence of work tasks – the workflow?
- What is the hierarchy of tasks?

Interface Design

Once interface analysis has been completed, all tasks required by the end-user have been identified in detail. Using information developed during interface analysis (Section 12.3), define interface objects and actions (operations).

Define events (user actions) that will cause the state of the user interface to change. Model this behavior.

Depict each interface state as it will actually look to the end-user. Indicate how the user interprets the state of the system from information provided through the interface.

Interface Design Patterns

The complete UI	Page layout
Forms and input	Tables
Direct data manipulation	Navigation
Searching	Page elements
E-Commerce	

Design Issues:

Response time	Help facilities
Error handling	Application accessibility
Internationalization	
Menu and command labeling	

Interface construction and implementation

The implementation activity starts with the development of prototype (model) that enables usage scenarios to be evaluated.

As iterative design process continues a User Interface toolkit that allows the creation of windows, menus, device interaction, error messages, commands, and many other elements of an interactive environment can be used for completing the construction of an interface.

Interface Validation

This phase focuses on testing the interface. The interface should be in such a way that it should be able to perform tasks correctly and it should be able to handle a variety of tasks.

It should achieve all the user's requirements. It should be easy to use and easy to learn. Users should accept the interface as a useful one in their work.

6.6 Interface Design Activities

There are a number of activities performed for designing user interface.

- GUI Requirement Gathering
- User analysis
- Task analysis
- GUI Design & implementation
- Testing

GUI Requirement Gathering

In this phase all functional and non-functional requirements are gathered for GUI. This can be taken from user and their existing software solution.

User analysis

The designer studies who is going to use the software GUI. The target audience matters as the design details change according to the knowledge and competency level of the user.

Task analysis

Designers have to analyze what task is to be done by the software solution.

Tasks can be represented in hierarchical manner taking one major task and dividing it further into smaller sub-tasks. Tasks provide goals for GUI presentation.

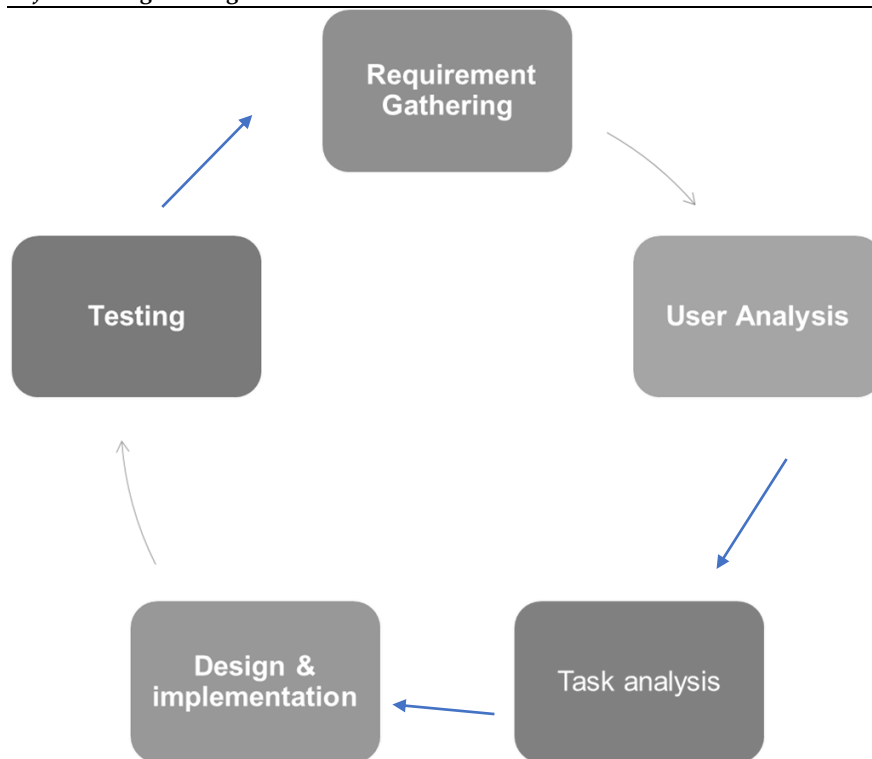
GUI Design & implementation

Designers after having information about requirements, tasks and user environment, design the GUI and implements into code and embed the GUI with working or dummy software in the background.

Testing

Testing can be done in various ways. Organization can have in-house inspection, direct involvement of users and release of beta version are few of them. Testing may include usability, compatibility, user acceptance etc.

Interface Design Activities



Summery

- User Interface Design is a process of designing or fabricating the interfaces through which the user can communicate with the computer.
- There are two types of User Interface
 - Command Line Interface
 - Graphical User Interface
- Golden rules stated by Theo Mandel that must be followed during the design of the interface.
 - Place the user in control
 - Reduce the users memory load
 - Make the interface consistent
- User interface Design Principles
 - Learnability
 - Flexibility
 - Robustness
 - Simplicity
 - Visibility

Four different models come into play when a user interface is to be analyzed and designed

- User model
- Design model
- Mental model
- Implementation model

Self Assessment

1. Which is part of user interface?
 - A. Command Line Interface
 - B. Graphical User Interface
 - C. Both Command Line Interface and Graphical User Interface
 - D. None of above

2. Which is not a part of Graphical User Interface?
 - A. Icon
 - B. Button
 - C. Command
 - D. Cursor

3. Command Line Interface is____
 - A. Minimal memory usage
 - B. Great for slow-running computers
 - C. This method relies primarily on the keyboard
 - D. All of above

4. Which is not a part of Command Line Interface?
 - A. UNIX
 - B. DOS
 - C. LINUX
 - D. None of above

5. Which is not a part of Graphical User Interface?
 - A. Windows XP
 - B. UNIX
 - C. LINUX
 - D. None of above

6. Graphical User Interface is____
 - A. Memorizing command lists is not necessary
 - B. Allows for running multiple applications, programs, and tasks simultaneously
 - C. Follow WYSIWYG (What You See Is What You Get)
 - D. All of above

7. Which is not a part of Interface Design Model?
 - A. User model
 - B. Design model
 - C. SDLC model
 - D. Mental model

8. How many phases in Interface design process?
 - A. 2
 - B. 4
 - C. 3
 - D. 5

9. Which of the following devices are mainly responsible for the user interface?
 - A. Input and output devices
 - B. Memory devices
 - C. Processor
 - D. None of the above

10. Which is not a part of Interface design process?
 - A. Interface design
 - B. Testing
 - C. Interface construction
 - D. User, task and environment analysis

Answer for Self Assessment

- | | | | | |
|------|------|------|------|-------|
| 1. C | 2. C | 3. D | 4. C | 5. B |
| 6. D | 7. C | 8. B | 9. A | 10. B |

Review Questions

1. What is significance of user interface design in software engineering?
2. Explain golden rules with suitable example.
3. What are the different elements of Graphical User Interface?
4. Discuss User interface Design Principles.
5. What are the Guidelines for user interface designing?
6. Discuss any two interface design models with example.
7. What are the four activities for interface design process?
8. Explain interface validation.

**Further Reading**

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